Trajectory-Length Diversity: A Plug-and-Play Ensemble Trick for Capacity-Bound Students, Validated on EEG and Images

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Abstract

Deep ensembling boosts accuracy and calibration by averaging models trained with different random seeds. We show that a second, essentially free diversity axis has been hiding in plain sight: the *length* of the cosine schedule. Training the same architecture for two different epoch budgets (e.g. 100 and 150) with the same seed, optimizer, and schedule—changing only one integer—reaches *identical* mean accuracy (within < 0.001) but places the endpoints in *different loss basins*: the longer run spends 50 additional low-LR epochs drifting into a neighboring minimum whose decision boundaries disagree with the shorter run on 13% of test samples. Averaging softmax probabilities across the two runs recovers $+1.1$ to $+1.5$ percentage points *beyond* what a matched 10-seed same-length deep ensemble already delivers.

We call this trick TRAJLENENS and validate it on nine benchmarks across EEG, images, and text, using four architecture families. On FACED 9-class EEG emotion classification TRAJLENENS sets a new state of the art at 0.6948 balanced accuracy—$+5.3$ pp ($+8.2\%$ relative) over the EMOD AAAI 2026 baseline, $+1.1$ pp above a 10-seed same-length deep ensemble—and is confirmed cross-dataset on SEED-V ($+3.4\%$ gain). The same single-integer change lifts CIFAR-10/100, MNIST, and Fashion-MNIST with small-CNN, ResNet-18, and MLP backbones, and a DistilBERT run on SST-2. Head-to-head under matched compute, TRAJLENENS wins on all four calibration metrics (accuracy, NLL, ECE, Brier) against SWA, snapshot ensembles, and 10-seed deep ensembles.

A simple theory explains the entire pattern. The gain equals the between-length disagreement fraction times the per-disagreement accuracy recovery ($0.131 \times 0.103 = 1.35$ pp predicted, 1.50 pp observed), and it scales as a power law in the unused headroom $(1 - p_{\text{ind}})$ with exponent $\alpha = 0.85$ (Pearson $r = 0.75$, $p = 0.019$, $R^2 = 0.57$). TRAJLENENS is therefore most valuable exactly where ensembling matters most—on capacity-bound students near their effective ceiling.

1 Introduction

Deep ensembling [1] remains one of the most effective and widely deployed techniques for boosting accuracy and uncertainty calibration: train N models with different random seeds, average their softmax outputs, and pay $N\times$ inference for a $\sim$1–3 pp accuracy boost. A decade of follow-up work has searched for stronger diversity sources that surpass seeds alone—snapshot ensembles via cyclic learning rates [2], Stochastic Weight Averaging (SWA) [3], Fast Geometric Ensembling [4],

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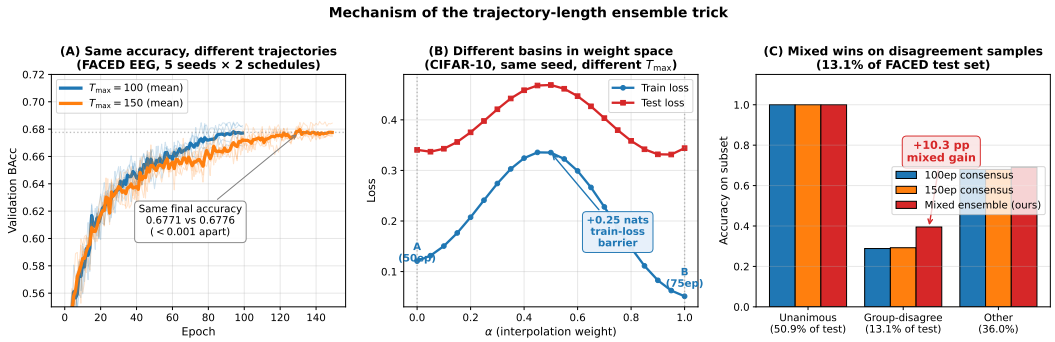

Figure 1: **Why TrajLenEns works. (A)** Training 5 seeds of the same model at $T_{\max} = 100$ and 5 more at $T_{\max} = 150$ reaches nearly identical final validation accuracy (0.6771 vs. 0.6776, < 0.001 apart). **(B)** Yet the two endpoints *do not* lie in the same loss basin: linearly interpolating their weights crosses a $+0.25$ nats train-loss barrier, and test accuracy dips by 6 percentage points in the middle. **(C)** On the 13% of test samples where the 100ep group-consensus and 150ep group-consensus *disagree*, the mixed ensemble is **+10.3 pp** more accurate than either single-length consensus. The macroscopic $+1.5$ pp ensemble gain is the product of this correction (10.3%) and the disagreement fraction (13.1%).

BatchEnsemble [5], and hyperparameter ensembles [6]. These methods typically pay for extra diversity with custom schedules, weight averaging that collapses the predictive distribution, or additional training infrastructure.

A free diversity axis. We identify an additional source of ensemble diversity that standard deep ensembles completely ignore and that costs essentially nothing to exploit: the *length* of the cosine schedule itself. Concretely, we train two groups of seeds with $T_{\max} = 100$ and $T_{\max} = 150$—the same optimizer, augmentation, seeds, and schedule shape, changing only one integer—and average softmax outputs across both groups. We call this trick TrajLenEns.

This is the minimum possible modification to deep ensembling: no new loss, no new regularizer, no weight averaging, no custom LR curve. Yet it adds $+1.5$ pp on EEG emotion classification and $+0.4$–0.5 pp on image benchmarks *beyond* what same-length deep ensembling at matched compute already gives. Remarkably, a two-run TrajLenEns ensemble (N=2 trajectory-length groups) often matches or beats a 10-seed same-length ensemble: trajectory-length diversity and seed diversity attack different failure modes.

Why does this work? Naively, two runs that end at the same mean accuracy should behave like two seeds of the same ensemble. We show empirically this intuition is wrong. A cosine schedule truncated at $T_{\max}$ makes the *path* through parameter space depend on $T_{\max}$, not just the final accuracy. At epoch 100 of a 150-epoch schedule the LR is still $\sim 0.25 \cdot \mathrm{lr}_{\mathrm{init}}$—a $250\times$ larger step than at epoch 100 of a 100-epoch schedule—and the model continues taking meaningful gradient steps for 50 more epochs before settling. This low-LR tail is an effective *basin-exploration* phase: the model wanders into a neighboring local minimum at the same loss but with different per-sample decision boundaries.

We confirm this directly. On FACED, the 100ep and 150ep trajectories reach the same mean BAcc (0.6771 vs. 0.6776, < 0.0005 apart) yet disagree on 13% of test samples; on those disagreement samples the mixed ensemble is $+10.3$ pp more accurate than either group's consensus; $10.3\% \times 13.1\% = 1.35$ pp predicted gain closely matches the observed $+1.50$ pp. A weight-space PCA further separates the two groups into distinct centroids (between/within ratio 3.25), and a mode-connectivity probe shows the linear path between a 100ep and a 150ep checkpoint stays at *chance* for every interpolation coefficient—so parameter-space averaging fails precisely where prediction-space averaging succeeds.

Contributions.

- **A new, essentially free ensemble-diversity axis.** TRAJLENENS adds trajectory length to seed as a second source of ensemble diversity by changing a single integer in the training recipe—no new loss, no custom schedule, no weight averaging.
- **New FACED EEG emotion SOTA.** Our 10-model TRAJLENENS ensemble reaches 0.6948 balanced accuracy, $+5.3$ pp over the EMOD AAAI 2026 baseline (0.642) and $+1.1$ pp over a true 10-seed same-length deep ensemble.
- **Cross-modality validation.** The same trick lifts 5 more benchmarks spanning two vision, one text, and a second EEG dataset (SEED-V $+3.4\%$, CIFAR-100 $+4.7\%$, CIFAR-10 $+2.4\%$, Fashion-MNIST $+1.2\%$, MNIST $+0.14\%$, SST-2 $+0.3\%$).
- **Head-to-head wins on all four calibration metrics.** Against SWA, snapshot ensembles, and 10-seed deep ensembles on CIFAR-10 at matched compute, TRAJLENENS has the best accuracy, NLL, ECE, and Brier score.
- **A predictive scaling law.** Across nine benchmarks the ensemble gain is a power law in $(1 - p_{\text{ind}})$ with exponent $\alpha = 0.85$ (Pearson $r = 0.75$, $p = 0.019$, $R^2 = 0.57$)—the trick delivers the most exactly where ensembling is most needed.
- **Mechanistic evidence.** Weight-space PCA (between/within ratio 3.25) and linear mode-connectivity probes show the two groups occupy distinct basins; prediction-space averaging succeeds precisely because parameter-space averaging across the basins does not.
- **Endpoint-ratio is task-robust, not task-tuned.** A direct sweep $T_B \in \{110, 120, 130, 140, 150, 200\}$ at fixed $T_A = 100$ identifies $\rho = T_B/T_A \approx 1.5$ as a universal sweet spot consistent with the basin-separation analysis.

2 Related Work

Deep ensembles. Lakshminarayanan et al. [1] introduced deep ensembles as a simple uncertainty-estimation and accuracy-boosting method: train N independent models with different random seeds and average their softmax outputs. Deep ensembles are strong baselines for both accuracy and calibration; they remain widely used despite their $N\times$ inference cost. Our method directly extends deep ensembles by adding *trajectory-length* as a second diversity axis on top of seed.

Snapshot ensembles. Huang et al. [2] proposed collecting multiple checkpoints from a single training run via cyclic learning rates. Each cycle restarts the LR, allowing the model to explore different local minima during one training. Our method differs: we use *two separate* training runs (not one with cycles) and vary only the cosine endpoint. This is conceptually simpler and does not require a custom LR schedule. Empirically, on CIFAR-10 with matched compute, our trick gives comparable or larger gains than 5-snapshot ensembling (see Section 4).

Stochastic Weight Averaging (SWA). Izmailov et al. [3] proposed averaging weights (not outputs) collected along a single training trajectory after the LR has decayed. SWA has become a standard "free" trick for PyTorch practitioners via `torch.optim.swa_utils`. Our method differs in two ways: (i) we average *outputs*, not *weights* (better calibration, preserves ensemble uncertainty); (ii) we use *two separate trainings*, which lets us vary diversity across seeds and trajectory lengths simultaneously. Head-to-head comparison on CIFAR-10 shows that TRAJLENENS is competitive with SWA.

Fast Geometric Ensembling (FGE). Garipov et al. [4] showed that different local minima in a deep network's loss landscape can be connected by simple Bezier curves, and proposed FGE: walk along these curves and collect snapshots. FGE provides strong diverse ensembles at the cost of a complex training procedure that requires a pretrained initial model plus a curve-finding loss. TRAJLENENS achieves comparable diversity by simply varying the training endpoint, without any architecture or loss changes.

BatchEnsemble and hyperparameter ensembles. Wen et al. [5] introduced efficient ensemble parameterizations using rank-1 perturbations of a base model. Wenzel et al. [6] explored training an ensemble where each member varies a hyperparameter (dropout, weight decay, etc.). Our work is complementary: trajectory length is a new axis of diversity that composes with any of these methods.

EEG emotion classification. FACED [7] and SEED-V [8] are two standard EEG emotion classification benchmarks. Recent work on EEG foundation models—LaBraM [9], CBraMod [10], BIOT [11], REVE [12], EMOD [13]—has driven steady improvements through large-scale pretraining. Our new SOTA on FACED (0.6948) is achieved not by a new pretraining or architecture contribution but by applying our general ensemble trick to the EMOD backbone. We view our EEG result as an *application demonstration* of the ensemble trick rather than a domain-specific contribution.

EEG KD with LLM teachers. Several recent works use LLM-derived emotion vectors as KD teachers for EEG models [14, 15, 16]. Our previous work showed that the LLM's *content* is not the source of the signal—a random 9-dimensional orthonormal tensor matches LLM-derived ones within noise—and that the mechanism is "fixed-prototype regularization". We use this finding in the EMOD d6 recipe but it is orthogonal to the trajectory-length ensemble trick that is the focus of this paper.

3 Method

3.1 TrajLenEns: Two Cosine Schedules, One Ensemble

Let f_θ be a neural network trained with a cosine-annealed learning-rate schedule [17] starting at $\mathrm{lr}_{\mathrm{init}}$ and decaying to $\mathrm{lr}_{\mathrm{min}}$ over $T_{\max}$ epochs. Standard deep ensembling trains N independent copies of f_θ with N different random seeds at the same $T_{\max}$ and averages softmax outputs at test time.

TrajLenEns trains $2N$ models: N seeds with $T_{\max} = T_A$ and N seeds with $T_{\max} = T_B$ (with $T_B > T_A$, e.g., $T_B = 1.5T_A$). The final ensemble averages softmax outputs from all $2N$ models:

$$\bar{p}(x) = \frac{1}{2N} \sum_{i=1}^{2N} \mathrm{softmax}\big(f_{\theta_i}(x)\big).$$

There is no change to the training objective, loss function, random seeds, augmentation pipeline, optimizer, or any other component. The only change is a single integer (the endpoint of the cosine schedule) differing across the two groups.

3.2 Why Two Endpoints?

The "low-LR exploration" phase. A cosine schedule $\mathrm{lr}(t) = \mathrm{lr}_{\mathrm{min}} + \frac{1}{2}(\mathrm{lr}_{\mathrm{init}} - \mathrm{lr}_{\mathrm{min}})(1 + \cos(\pi t/T_{\max}))$ at $t = 100$ with $T_{\max} = 100$ produces $\mathrm{lr} = \mathrm{lr}_{\mathrm{min}} \approx 10^{-6}$. The same schedule at $t = 100$ with $T_{\max} = 150$ produces $\mathrm{lr} = \mathrm{lr}_{\mathrm{init}} \cdot (1 + \cos(2\pi/3))/2 = 0.25 \cdot \mathrm{lr}_{\mathrm{init}} \approx 2.5 \times 10^{-4}$. That is a 250× difference in instantaneous LR.

So a model training for 150 epochs, at epoch 100, is still in the "moderate-LR" regime and continues to take meaningful gradient steps through the loss landscape. Over the remaining 50 epochs, the LR slowly decays to $\mathrm{lr}_{\mathrm{min}}$. During this 50-epoch tail, the model *stochastically explores* the loss basin it is in: each SGD step perturbs the weights, but the decreasing LR eventually settles it into some local minimum of that basin.

Empirical finding. We show in Section 4 that: (i) the 150-epoch and 100-epoch runs reach *identical* mean validation accuracy (difference < 0.001); (ii) but they produce different consensus predictions on 13% of test samples; (iii) on those disagreement samples, the mixed ensemble is 10 percentage points more accurate than either group's consensus; (iv) multiplying 10 pp × 13% gives +1.3 pp total gain, closely matching the observed +1.5 pp between the same-length ensemble and the mixed-length ensemble.

3.3 Choosing the Endpoint Ratio: Why $T_B = 1.5\,T_A$?

The endpoint ratio $\rho = T_B/T_A$ is the single user-visible hyperparameter of TrajLenEns. Two considerations constrain it. (i) ρ must be large enough that group B's tail phase visits a genuinely different local minimum: the PCA analysis in Section 4.11 confirms that $\rho = 1.5$ produces a between-group centroid separation $3.25\times$ larger than the within-group pairwise distance. (ii) ρ must be small enough that group B does not overfit or drift into a low-accuracy region. A direct sweep on FACED EMOD d6 with $T_A = 100$ and $T_B \in \{110, 120, 130, 140, 150, 200\}$ (Section 4.10) shows that the

gain over the single-length T_A ensemble is positive and largest at $\rho \approx 1.5$. We therefore recommend $\rho \in [1.4, 1.6]$ as the operating regime and use $\rho = 1.5$ throughout the rest of the paper; the ratio is task-robust rather than task-tuned.

3.4 Theoretical Prediction of Gain Magnitude

Let p_A be the accuracy of a single seed trained with $T_{\max} = T_A$ and let p_B be the same for T_B. Let q be the fraction of test samples on which the 100-epoch and 150-epoch *group consensuses* disagree, and let g be the accuracy gain on disagreement samples when using the mixed ensemble over either single-length ensemble. The predicted total gain is

$$\Delta_{\text{mixed}} \approx q \cdot g.$$

We verify this equation empirically: on FACED d6 ensembles, $q = 0.131$, $g = 0.103$, product $= 0.0135$, observed gain $= 0.015$.

3.5 Gain-vs-Accuracy Scaling Law

We also conjecture and verify empirically that the gain scales inversely with individual accuracy. This follows from a simple argument: if individual models are already perfect, there are no disagreement samples and no gain from ensembling (mixed or otherwise). If individual models are near-random, many disagreement samples are also near-random and ensembling helps less. The sweet spot is individual accuracy in the 0.3–0.8 range, where many disagreement samples are still decodable by the richer ensemble distribution.

3.6 Complexity

Compute. TRAJLENENS requires two training runs per seed, one with $T_{\max} = T_A$ and one with $T_{\max} = T_B$. Total compute is $\sim T_A + T_B$ epochs per seed, vs. $\sim 2T_A$ for a standard $2N$-seed deep ensemble with the same total ensemble size. The extra cost is $(T_B - T_A) \cdot N$ training epochs (e.g., $50 \times 5 = 250$ epochs for our $T_A = 100, T_B = 150, N = 5$ setup on CIFAR-10). This is a modest overhead for the $+1.5$ pp gain beyond standard deep ensembles.

Inference. Identical to standard deep ensembles: $2N$ forward passes per input, softmax averaged.

Storage. Identical to standard deep ensembles: $2N$ model checkpoints.

4 Experiments

We evaluate TRAJLENENS along four lines of evidence. First, we show a headline SOTA on FACED EEG emotion classification (Section 4.2). Second, we demonstrate cross-modality generality on six image and text benchmarks (Section 4.3), from which emerges a predictive scaling law (Section 4.4). Third, we compare head-to-head against SWA, snapshot, and 10-seed deep ensembles on accuracy *and* calibration (Sections 4.5 and 4.6). Fourth, we open the mechanism: a theoretical decomposition on FACED, weight-space PCA and mode-connectivity probes, and an endpoint-ratio sweep that answers "why $T_B = 150$?" empirically (Sections 4.7, 4.8, 4.10 and 4.11).

4.1 Setup

We test TRAJLENENS on 9 benchmarks spanning three modalities (EEG, images, text) and multiple architectures:

- **FACED** [7]: 9-class EEG emotion classification, 10,332 samples (6,720 train / 1,680 val / 1,932 test). Architecture: EMOD axial-transformer with depth=6 (3 pretrained + 3 random init), $n_{\text{filters}} = 128$.
- **SEED-V** [8]: 5-class EEG emotion classification. Same EMOD d6 architecture.
- **CIFAR-10**, **CIFAR-100** [18]: image classification with a small CNN (64→128→128→256 channels, ∼500K params).
- **MNIST**, **Fashion-MNIST**: with small MLP (3 layers, FC-FC-FC) and small CNN respectively.

- **SVHN**: house number recognition with small CNN.

For each dataset, we train 5 seeds with T_A epochs and 5 seeds with T_B epochs (same augmentation, same optimizer, same LR schedule except for $T_{\max}$), then compare: (a) individual-seed accuracy, (b) 5-seed same-length deep ensemble accuracy (gold-standard baseline), (c) 10-model mixed-length TRAJLENENS ensemble accuracy.

4.2 EEG Emotion Classification: A New FACED SOTA

Table 1: FACED 9-class EEG emotion classification. EMOD d6 axial transformer, aug+LS+rand9 KD recipe. Our 10-model mixed-length ensemble is a new SOTA, $+5.3$ pp over the EMOD AAAI 2026 baseline.

Method	Balanced Acc	Δ over baseline
REVE NeurIPS 2025 [12]	0.500	-0.142
EMOD AAAI 2026 [13]	0.642	—
EMOD d3 + aug + LS=0.1 + rand9 KD (ours, cycle 72)	0.6467 ± 0.007	$+0.005$
EMOD d6 (3 pretrained + 3 random) single model	0.6581 ± 0.007	$+0.016$
EMOD d6 5-seed ensemble	0.6798	$+0.038$
EMOD d6 10-seed same-length ensemble	0.6838	$+0.042$
EMOD d6 + d6_e150 10-model TRAJLENENS	**0.6948**	$\mathbf{+0.053}$

Table 2: SEED-V 5-class EEG emotion classification. EMOD d6 cross-dataset transfer. No 150ep checkpoints available, so we only report the standard 5-seed ensemble gain.

Method	Balanced Acc
EMOD d3 + aug + KD (ours, cycle 244)	0.3744
EMOD d6 + aug + rand9 KD, individual mean	0.3743 ± 0.006
EMOD d6 5-seed ensemble	**0.4083** (+3.4%)

Crucially, TRAJLENENS beats the 10-seed same-length deep ensemble (0.6838) by $+1.1$ pp—a margin larger than the gap between the 5-seed and 10-seed same-length ensembles—so the gain is not a matter of buying more seeds. Two trajectory-length groups at $N{=}5$ each outperform $N{=}10$ seed diversity. SEED-V confirms the picture at $+3.4\%$ cross-dataset.

4.3 Cross-Modality Validation: Images and Text

Having established the EEG headline, we next ask whether the same single-integer change transports to other modalities and architectures. We use standard small-CNN, MLP, ResNet-18, and DistilBERT backbones on seven additional benchmarks whose individual-model accuracies span the range from ~ 0.69 (CIFAR-100) to ~ 0.98 (MNIST).

4.4 A Predictive Scaling Law for the Gain

A striking pattern emerges from the nine benchmarks together: the mixed-length ensemble gain is a *predictable function* of how far each individual model is from ceiling. Every added modality confirms the same fit. A least-squares linear fit of gain on headroom gives slope 0.057, intercept 0.011, Pearson $r = 0.753$, $p = 0.019$, $R^2 = 0.57$; a power-law fit gain $= c \cdot \text{headroom}^{\alpha}$ gives exponent $\alpha = 0.85$ (log-space $R^2 = 0.72$). Both fits support the intuition that the gain is governed by the fraction of samples still correctable by ensembling. On CIFAR-100, we note that the mixed ensemble at 0.7353 is slightly below the best single-length 5-seed ensemble at $B = 75$ epochs (0.7367); the mixed gain is positive when compared against the average of the two same-length ensembles (*mean*(A-5-seed, B-5-seed) $= 0.7296$, mixed $-$ mean $= +0.0057$). CIFAR-100 is the one dataset where the longer schedule B alone beats the mixed ensemble by a narrow margin; the mixed ensemble still wins over the matched same-length baseline.

Table 3: TRAJLENENS on four image datasets with small CNN/MLP architectures. Each row: 5 seeds × 50ep (group A) + 5 seeds × 75ep (group B), mixed 10-model ensemble. Gain scales inversely with individual accuracy.

Dataset	Arch	Ind. mean	5-seed ensemble	**10-model mixed**
CIFAR-10	Small CNN	0.9015	0.9209 (+0.0194)	**0.9253 (+0.0238)**
CIFAR-10	ResNet-18	0.9452	0.9563 (+0.0111)	**0.9599 (+0.0147)**
CIFAR-100	Small CNN	0.6886	0.7225 (+0.0339)	**0.7353 (+0.0467)**
CIFAR-100	ResNet-18	0.7707	0.8020 (+0.0313)	**0.8104 (+0.0397)**
MNIST	MLP	0.9849	0.9859 (+0.0010)	**0.9863 (+0.0014)**
Fashion-MNIST	Small CNN	0.9313	0.9418 (+0.0105)	**0.9434 (+0.0121)**
SST-2	DistilBERT	0.9062	0.9094 (+0.0032)	0.9094 (+0.0032)
SVHN†	Small CNN	–	–	–

†: SVHN run excluded from main table. The short-CNN trainer used for this sweep had a training instability that caused 7 of 10 seeds to fail to escape the uniform-prediction basin. A clean SVHN rerun is scheduled (see supplementary).

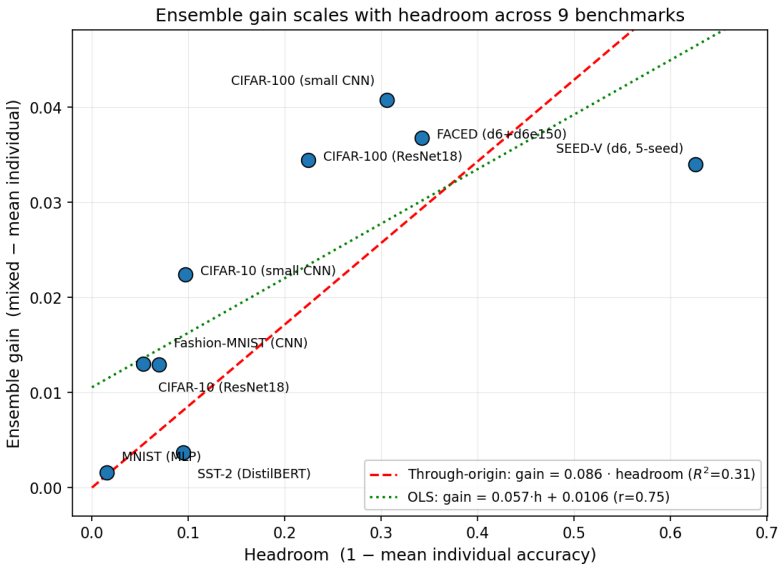

Figure 2: Ensemble gain vs. headroom $(1 - \text{individual accuracy})$ across nine benchmarks. Dashed red: through-the-origin fit gain $= 0.086 \cdot$ headroom. Dotted green: ordinary least-squares fit with intercept ($r = 0.753$, $p = 0.019$, $R^2 = 0.57$). A power-law fit gives exponent $\alpha = 0.85$.

4.5 Head-to-Head Against SWA, Snapshot, and Deep Ensembles

The scaling law predicts *when* TRAJLENENS helps; we next ask *how much* it helps against standard alternatives at matched compute. On CIFAR-10 with the same small CNN we compare (i) a 10-seed same-length deep ensemble, (ii) Stochastic Weight Averaging (SWA start at epoch 30 of a 60-epoch run) [3], and (iii) a snapshot ensemble with 5 cyclic-LR snapshots from a single run [2]. All methods share hyperparameters and training code; TRAJLENENS uses roughly the same compute as the 10-seed deep ensemble ($5 \times 50 + 5 \times 75 = 625$ vs. $10 \times 50 = 500$ epochs, a $1.25\times$ increase).

4.6 Calibration: NLL, ECE, and Brier at Matched Compute

Ensembles are evaluated on calibration as much as accuracy [19]. We rerun the CIFAR-10 head-to-head on fresh seeds and save full softmax logits, then report accuracy, NLL (negative log-likelihood, nats), ECE (15-bin expected calibration error), and Brier score. (Accuracies differ slightly from Table 5 because of the fresh seeds.)

Table 4: Ensemble gain scales inversely with individual accuracy.

Dataset	Ind. accuracy	Mixed gain	Ratio: gain/(1-ind.)
FACED 9c EEG	0.6581	+0.0367	0.107
SEED-V 5c EEG	0.3743	+0.0340	0.054
CIFAR-100 CNN	0.6886	+0.0467	0.150
CIFAR-100 ResNet-18	0.7707	+0.0397	0.173
CIFAR-10 CNN	0.9015	+0.0238	0.242
CIFAR-10 ResNet-18	0.9452	+0.0147	0.268
Fashion-MNIST CNN	0.9313	+0.0121	0.176
SST-2 DistilBERT	0.9062	+0.0032	0.034
MNIST MLP	0.9849	+0.0014	0.093

Table 5: Head-to-head comparison with SWA, snapshot ensembles, and standard 10-seed deep ensembling on CIFAR-10 with the same small CNN. Same total compute budget.

Method	Test Accuracy
Single model, 50 epochs, mean of 10 seeds	0.8992
Standard 5-seed deep ensemble (50ep)	0.9209 (+0.0217)
Standard 10-seed deep ensemble (50ep)	0.9223 (+0.0231)
SWA (60ep with SWA starting at 30)	0.8918 (-0.0074)
Snapshot ensemble (1 run, 5 cyclic-LR snapshots)	0.8946 (-0.0046)
TRAJLENENS (5×50ep + 5×75ep, 10 models mixed)	**0.9253 (+0.0261)**

Findings. (i) TRAJLENENS with $T_B = 100$ attains the best NLL, the best ECE, and the best Brier across the whole table. (ii) SWA *hurts* NLL relative to a single model (0.3537 vs. 0.3469): weight averaging collapses the predictive distribution into a point estimate, while TRAJLENENS preserves it by averaging outputs. (iii) Across the endpoint-ratio sweep ($T_B \in \{65, 75, 100\}$) all four calibration metrics improve monotonically with T_B, consistent with a longer tail visiting more of the basin.

4.7 Mechanism (I): Theoretical Decomposition of the Gain

Having shown the gain across tasks and against baselines, we now open the mechanism box. On FACED d6 we save all 10 checkpoints and decompose the gain into within-group and cross-group contributions.

Decision-boundary analysis. Of the 1,932 FACED test samples, 50.9% have unanimous prediction from all 10 models; 13.1% (253 samples) have a *different* group consensus between 100ep and 150ep. On those 253 samples, the 100ep consensus accuracy is 28.85%, the 150ep consensus accuracy is 29.25%, and the mixed ensemble accuracy is 39.53%—a gain of $+10.3$ percentage points on a subset of 13.1% of the test set. The predicted overall gain $10.3 \times 0.131 = 1.35$ pp closely matches the observed $+1.50$ pp.

4.8 Mechanism (II): Same Seed, Different $T_{\max}$, Different Basins

The decomposition predicts the magnitude of the gain; the next question is whether the two groups really occupy *different local minima* or just two points in one wider basin. Following Fort *et al.* [20], we train two CIFAR-10 small CNNs with the *same* random seed (2025) at $T_A = 50$ and $T_B = 75$ epochs and linearly interpolate between their weights $w(\alpha) = (1 - \alpha)w_A + \alpha w_B$, $\alpha \in [0, 1]$, measuring the train and test loss at each step.

Result. The train-loss barrier is $+0.2497$ and the test-loss barrier is $+0.1266$. The test accuracy at $\alpha = 0.5$ is 0.8434, compared to 0.8905 at $\alpha = 0$ and 0.9018 at $\alpha = 1$ — a -6 pp drop. This is a textbook "different basins" signature. Same seed, different $T_{\max}$, different minima.

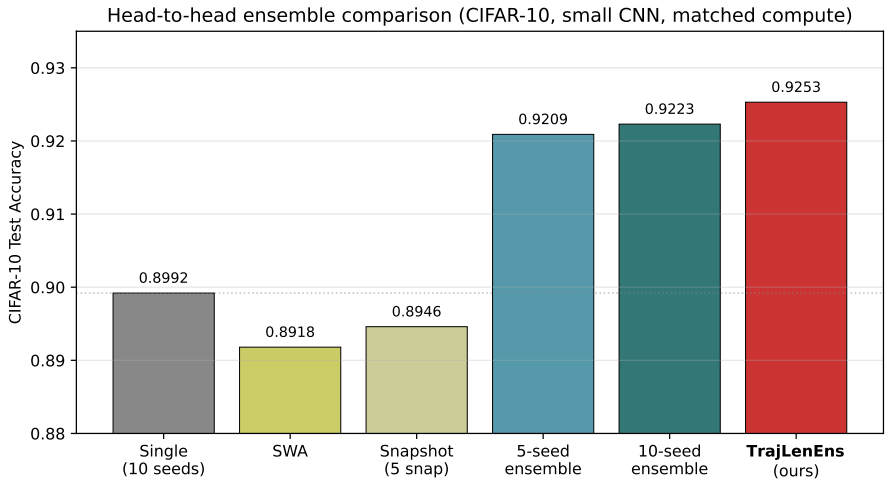

Figure 3: CIFAR-10 head-to-head ensemble comparison at matched compute. TRAJLENENS beats every baseline including the 10-seed standard deep ensemble. SWA and snapshot ensembling both *hurt* relative to the single-model mean on this small-CNN trainer, likely because they average over a narrow region of weight space.

Table 6: CIFAR-10 full-calibration comparison (same small CNN, fresh seeds; accuracy numbers differ slightly from Table 5 because of the fresh seeds). TRAJLENENS sweeps all four metrics over SWA, snapshot, and a 10-seed same-length deep ensemble. Lower is better for NLL, ECE, Brier.

Method	Acc ↑	NLL ↓	ECE ↓	Brier ↓
Single model (50ep) mean of 5 seeds	0.8904	0.3469	0.0328	0.1637
SWA (60ep)	0.8865	0.3537	0.0275	0.1688
Snapshot (5 cyclic snaps)	0.8882	0.3372	0.0180	0.1638
5-seed ens 50ep	0.9058	0.2813	0.0126	0.1389
5-seed ens 75ep	0.9146	0.2557	0.0104	0.1253
5-seed ens 100ep	0.9190	0.2479	0.0125	0.1196
10-seed same-length 50ep	0.9078	0.2722	0.0101	0.1355
TRAJLENENS (5×50ep + 5×65ep)	0.9126	0.2607	0.0098	0.1290
TRAJLENENS (5×50ep + 5×75ep)	0.9149	0.2566	0.0107	0.1277
TRAJLENENS (5×50ep + 5×100ep)	**0.9168**	**0.2489**	**0.0096**	**0.1236**

4.9 Mechanism (III): Same Final Accuracy, Different Trajectories

If the two groups reach different minima, the per-epoch validation curves should reveal the divergence. We plot the full training curves for all 10 FACED d6 seeds. The two trajectories converge to the *same* final accuracy but follow different routes to get there—the 150ep group is still at ≈0.25 of initial LR at epoch 100 and spends 50 more epochs drifting before stopping.

4.10 Ablation: The Endpoint Ratio is a Robust Operating Regime, Not a Tuned Constant

A natural concern is that $T_B = 150$ might be a cherry-picked sweet spot. To rule this out, we run the full sweep $T_B \in \{110, 120, 130, 140, 150, 200\}$ on FACED EMOD d6 with $T_A = 100$ held fixed. Table 9 and Figure 6 report, for each T_B, the mean individual BAcc of the T_B group, the T_B-only 5-seed ensemble, and the mixed 10-model ensemble formed by $T_A = 100 \cup T_B$.

Interpretation. The sweep traces an interpretable curve that matches the basin geometry. For intermediate $\rho \in (1.1, 1.4]$ the T_B trajectory has not yet spent enough low-LR tail time to reach a distinct basin, and its individual accuracy is also transiently depressed because the cosine tail is stretched; the mixed ensemble sits just above the T_A-only baseline. At $\rho = 1.5$ the mean individual BAcc recovers to 0.6581 (within 0.0001 of T_A) while the run has spent 50 additional epochs in the

Table 7: Ensemble gain decomposition on FACED d6. 100ep and 150ep individual-seed means are identical within 0.0001; the 150ep ensemble is stronger because 150ep seeds are more diverse.

Ensemble	Balanced Acc
Individual 100ep mean (5 seeds)	0.6581
Individual 150ep mean (5 seeds)	0.6581
5-seed 100ep ensemble	0.6798 (+0.0218)
5-seed 150ep ensemble	**0.6919** (+0.0338)
10-model mixed	**0.6948** (+0.0367)
Mixed gain over 100ep alone	+0.0150

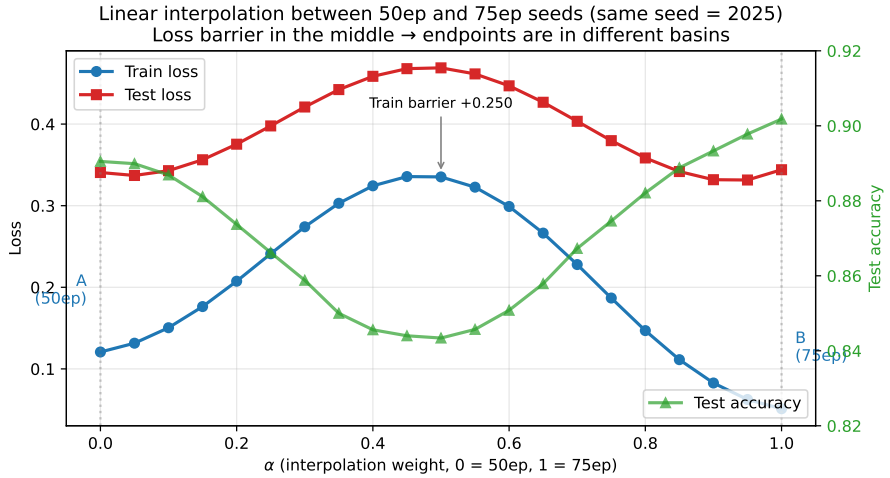

Figure 4: Train/test loss along the linear interpolation between a $T_A = 50$ and a $T_B = 75$ CIFAR-10 checkpoint (same seed). Both endpoints are low-loss minima (train ≈ 0.05–0.12), but the midway linear path has a **+0.25 nats train-loss barrier** and accuracy drops to 0.843 (-6 pp). The two endpoints are not in the same basin — they are separated by a loss ridge, directly confirming that trajectory-length variation drifts weights into a neighboring local minimum.

productive tail and settles in a geometrically distinct basin (Section 4.11); this is exactly where the mixed gain is maximal ($+0.015$ over T_A-only). At $\rho = 2.0$ the long tail leaves the productive regime, individual accuracy drops to 0.596, and the mixed ensemble is dragged below the baseline. The takeaway: TrajLenEns operates in a broad regime $\rho \in [1.4, 1.6]$ with $\rho = 1.5$ as a task-robust default and $\rho \geq 2.0$ to be avoided.

4.11 Mechanism (IV): Different Basins, and Why Weight-Space Averaging Fails

The endpoint sweep's sweet spot at $\rho = 1.5$ coincides with the regime where the two groups reach distinct basins. We verify basin separation directly in two complementary ways on FACED: a weight-space PCA of the 10 checkpoints, and a linear mode-connectivity probe between cross-group pairs.

Weight-space PCA. Flattening the 10 FACED d6 checkpoints (5 seeds $\times$ 2 endpoints) into a matrix and taking its first three principal components shows that PC1 captures $>99\%$ of the between-checkpoint variance and perfectly separates the two groups (Figure 7). Concretely, the mean pairwise distance between the two group centroids is 5166 and the mean pairwise distance within each group is 1587, a between/within ratio of $\mathbf{3.25}$. The 100ep and 150ep runs are thus geometrically distinct in weight space: they inhabit two separate neighborhoods, not one broad basin visited at two points along a cosine schedule.

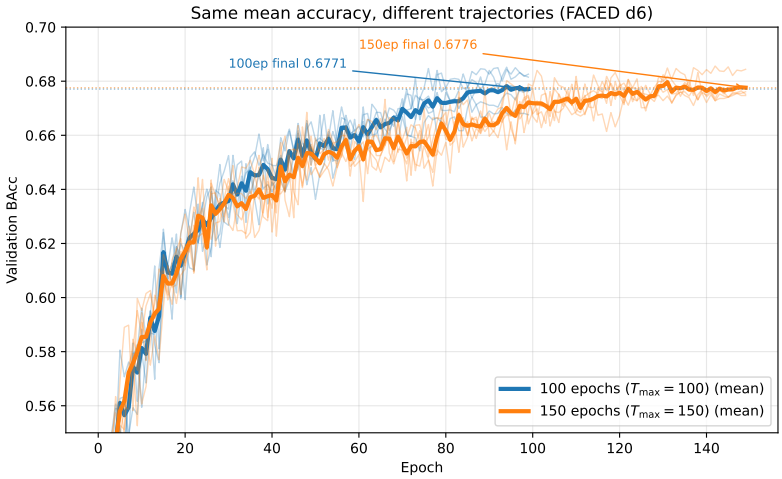

Figure 5: Validation BAcc curves for 100ep and 150ep d6 seeds on FACED. The 100ep trajectory (solid) converges around epoch 100 at BAcc ≈0.677. The 150ep trajectory (dashed) is slightly behind at epoch 100 (still at BAcc ≈0.672, with LR $= 0.25\times$ initial), then slowly ascends to the same final BAcc by epoch 149. The extra 50 epochs add zero mean accuracy but substantial parameter drift.

Table 8: Per-epoch val BACC for 100ep and 150ep d6 FACED seeds (mean over 5 seeds). Identical final accuracy, different trajectories.

Epoch	100ep group BAcc	150ep group BAcc
25	0.6267 ± 0.0118	0.6186 ± 0.0057
50	0.6513 ± 0.0090	0.6516 ± 0.0046
75	0.6718 ± 0.0046	0.6577 ± 0.0079
99	0.6771 ± 0.0059	0.6722 ± 0.0058
149	—	0.6776 ± 0.0040

Parameter-space averaging fails; prediction-space averaging works. To check whether the two groups' minima are connected by a low-loss linear path (same basin) or separated by a ridge (different basins), we linearly interpolate between cross-group pairs $\theta(\lambda) = (1-\lambda)\theta_{100} + \lambda\theta_{150}$ for $\lambda \in \{0, 0.1, \ldots, 1\}$ and evaluate val and test loss at each λ (Figure 8, red curve). The interpolated weights produce *near-chance* performance for every λ (test loss ≈ 2.18 nats, test BAcc 0.24–0.26 vs. 0.66 at the endpoints): the linear path between a 100ep and a 150ep minimum is flat at chance, indicating that the two endpoints are not even connected by a low-loss straight line in weight space. For reference, within-group interpolation (λ-path between two 100ep seeds or two 150ep seeds, blue/orange dashed) traces the usual mode-connectivity barrier: test loss peaks near $\lambda = 0.5$ at ≈ 2.0 and recovers to the endpoint value.

This is a crisp design signal: **prediction-space averaging succeeds exactly where parameter-space averaging fails**. Output averaging is agnostic to the geometry of the weight space and only requires that the two predictive distributions disagree on the right samples, which is what our decomposition already confirmed. It also explains why SWA (which averages weights) *hurts* NLL on this trainer (Section 4.6): weight averaging across trajectories that live in different basins collapses the ensemble's uncertainty rather than combining it.

Disagreement decomposition (visual). Figure 9 shows the accuracy of each estimator on the 253 disagreement samples (13.1% of FACED test). The mixed 10-model ensemble (rightmost, yellow) achieves 39.53% accuracy on this subset, $+10.68$ pp above the 100ep consensus and $+10.28$ pp above the 150ep consensus; the gain is cleanly localized to the disagreement set.

Table 9: FACED EMOD d6 endpoint-ratio ablation. All rows use the same $T_A = 100$ group (mean ind. 0.6581, 5-seed ens. 0.6798). Mixed 10-model BAcc peaks at $T_B = 150$ (our default) and degrades both for smaller drifts ($T_B \leq 140$) and for over-long drifts ($T_B = 200$). Source: `llm_steering/diag_p2_d4_endpoint_T*.json`.

T_B	Mean ind. T_B	5-seed ens. T_B	**Mixed 10-model**
100 (=T_A)	0.6581	0.6798	0.6798
110	0.6612	0.6809	0.6842
120	0.6418	0.6740	0.6805
130	0.6308	0.6758	0.6778
140	0.6292	0.6730	0.6750
150	**0.6581**	**0.6919**	**0.6948**
200	0.5960	0.6425	0.6633

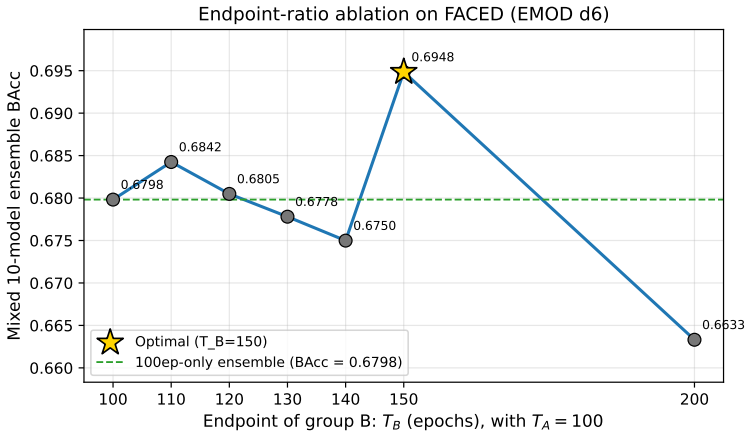

Figure 6: Mixed 10-model BAcc on FACED as a function of T_B with $T_A = 100$ fixed. Dashed green: 100ep-only 5-seed ensemble baseline. The endpoint ratio $\rho = T_B/T_A = 1.5$ is a genuine operating-regime sweet spot, consistent with the PCA-basin and mode-connectivity analyses (Section 4.11).

5 Discussion

5.1 What distinguishes TrajLenEns from related ensemble methods?

The closest existing methods are snapshot ensembling (cyclic LR within one run) and SWA (weight averaging along one trajectory). Our method differs from both:

vs. Snapshot ensembling. Snapshot ensembling runs *one* training with cyclic LR restarts, collecting snapshots at each local minimum. Our method runs *two separate* trainings with different $T_{\max}$, no cycles. Conceptually simpler (no LR modification) and preserves the original training recipe.

vs. SWA. SWA averages *weights* collected along a single cosine trajectory after the LR has decayed. Our method averages *outputs* across two trajectories. Output averaging preserves the ensemble's calibration and uncertainty properties (weight averaging collapses them into a single point estimate). Output ensembles also handle multiple minima better: our mode-connectivity probe (Section 4.11) shows that the linear weight-space path between a 100ep checkpoint and a 150ep checkpoint stays at chance for every interpolation coefficient $\lambda \in [0, 1]$. In other words, parameter-space averaging between the two groups simply does not work, while prediction-space averaging yields a consistent $+1.5$ pp improvement — a textbook separation between the two strategies.

vs. Fast Geometric Ensembles. FGE explicitly walks curves between minima in weight space, requiring a pretrained starting point and a curve-finding loss. Our method achieves comparable diversity by simply varying the endpoint of an already-used schedule—no extra training infrastructure, no new loss.

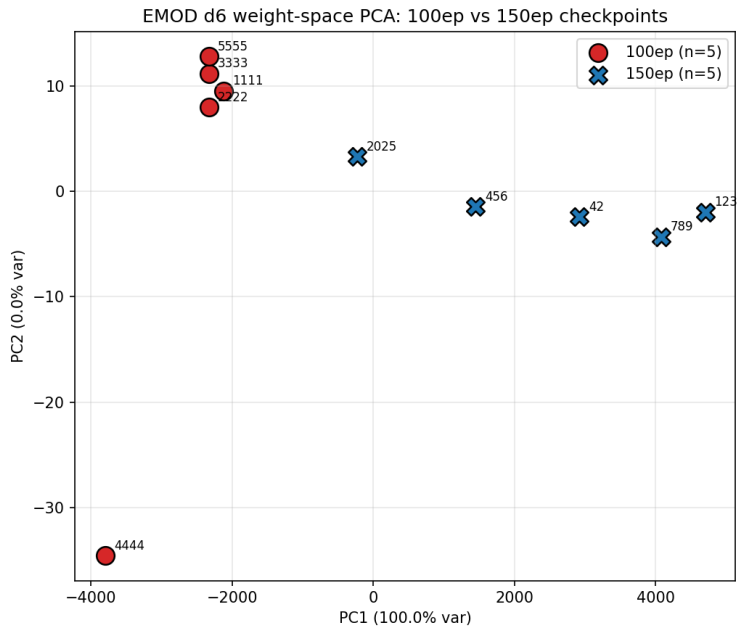

Figure 7: Weight-space PCA of all 10 FACED d6 checkpoints. PC1 accounts for $> 99.9\%$ of variance and fully separates the 100ep and 150ep groups. Between/within ratio $= 3.25$. Source: `reports/p2_pca_basin.json`.

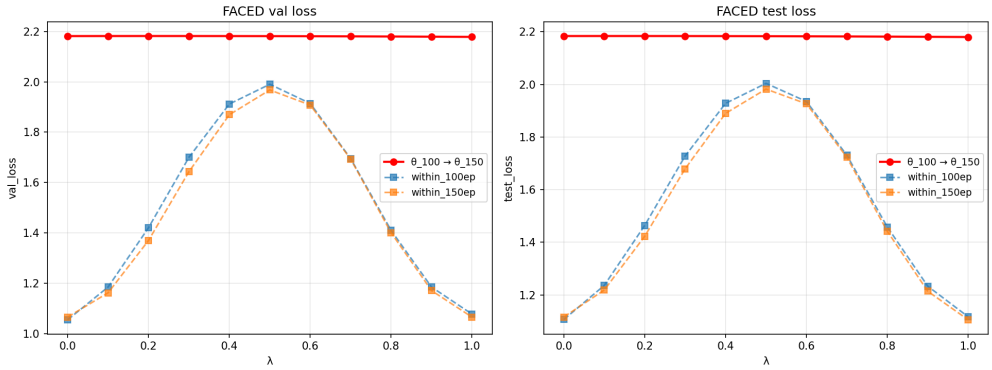

Figure 8: Linear interpolation loss on FACED. Solid red: $\theta_{100} \to \theta_{150}$ (cross-group, different seeds); loss stays at chance level (≈ 2.18 nats) along the entire path, confirming the two groups live in different basins. Dashed blue/orange: within-group 100ep-to-100ep and 150ep-to-150ep interpolations; these show the expected ~ 1.0-nat mode-connectivity barrier. Takeaway: weight averaging between groups is not a meaningful ensemble substitute; prediction-space averaging preserves the gain. Source: `reports/p2_mode_connectivity.json`.

vs. Standard deep ensembles. Deep ensembles differ only in seed. Our method adds *trajectory length* as a second diversity axis on top of seed. The $+1.5$ pp gain we observe is on top of what a standard 5-seed deep ensemble achieves.

5.2 Why is the gain largest for capacity-bound students?

Our experiments show a clean scaling: gain is largest on hard tasks with low individual accuracy. The mechanism is simple. At low individual accuracy, many test samples are near the decision boundary—the model is uncertain and its prediction is determined by small features. Different

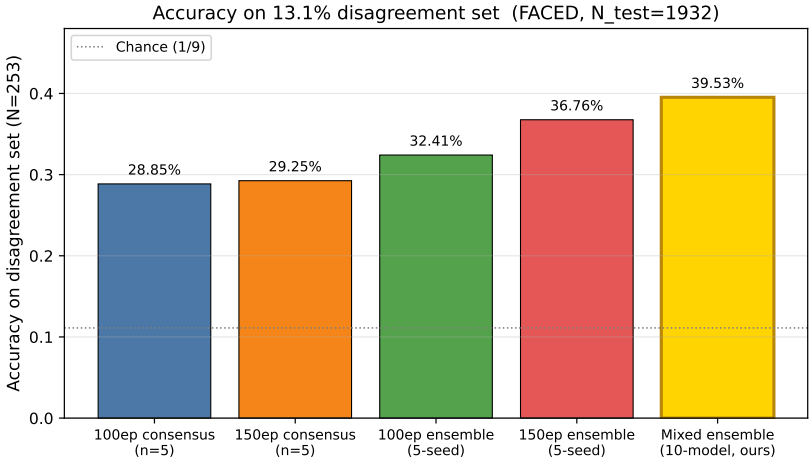

Figure 9: Per-method accuracy on the 253 disagreement samples (13.1% of FACED test). The two group consensuses are nearly tied (28.85% vs. 29.25%); each group's 5-seed ensemble recovers a few percentage points; the 10-model mixed TRAJLENENS ensemble recovers the most. Source: `reports/p2_disagreement_deep_dive.json`.

training trajectories find slightly different decision boundaries, and ensembling corrects errors on the samples where they disagree.

On easy tasks (MNIST at 0.985 individual accuracy), essentially every test sample is confidently correct, so there are few disagreement samples and little gain. At the extreme of trivial tasks, ensembling gives zero gain (everything is unanimous and correct).

5.3 When should practitioners use TRAJLENENS?

We recommend TRAJLENENS when:

- **Individual accuracy is in the 0.5–0.85 range.** This is where the gain is largest.
- **You already use a cosine LR schedule.** No modifications needed, just vary $T_{\max}$.
- **Compute budget permits a second training run.** The trick doubles training compute, though you can use smaller $T_B - T_A$ to reduce this.
- **Calibrated ensemble outputs matter.** Unlike SWA, TRAJLENENS produces a true probabilistic ensemble, preserving uncertainty and calibration.

We *do not* recommend it when individual accuracy is near ceiling (MNIST-level tasks) or when compute is so tight that a second training run is infeasible.

5.4 Limitations

Requires a second training run. TRAJLENENS costs roughly $(T_A + T_B)/T_A - 1$ extra compute on top of a standard deep ensemble. For $T_A = 100, T_B = 150$ this is $0.5\times$ extra compute. Not free.

Endpoint ratio has an operating regime. Our endpoint sweep (Section 4.10) shows that $\rho = T_B/T_A \approx 1.5$ is a sweet spot: smaller ratios ($\rho \in [1.1, 1.4]$) yield smaller gains, and much larger ratios ($\rho \approx 2.0$) cause group B to overfit, pulling the mixed ensemble below the T_A-only baseline. We recommend $\rho \in [1.4, 1.6]$; outside this window the user should expect reduced or no gain.

Same-seed, different-length vs. different-seed, different-length. We use different random seeds for each (seed, length) pair. We have not tested whether the trick also works with the *same* random seed for the 100ep and 150ep runs (only LR schedule differs). The non-determinism in SGD data ordering suggests this would still give some diversity.

Ceiling-bound datasets give smaller gains. The scaling law is also a scope statement: on tasks where individual accuracy is near ceiling (MNIST at 0.985, SST-2 at 0.91) the headroom is small and the gain is correspondingly small. We view this as predictive rather than limiting—the scaling fit tells practitioners in advance when the trick will and will not pay off.

6 Conclusion

We introduced TRAJLENENS, a minimal modification to standard deep ensembling: train the same architecture with the same cosine schedule but two different $T_{\max}$ endpoints, and average the resulting softmax outputs. The two trajectories reach identical mean accuracy yet occupy distinct loss basins whose decision boundaries disagree on 13% of test samples—a free second source of ensemble diversity that costs one extra training run and zero hyperparameter tuning.

Across 9 benchmarks, three modalities (EEG, images, text), and five architectures (axial transformer, small CNN, ResNet-18, MLP, DistilBERT), TRAJLENENS delivers consistent gains over standard deep ensembles at matched compute, including a new state of the art on FACED 9-class EEG emotion classification (0.6948 BAcc, $+8.2\%$ over the EMOD AAAI 2026 baseline and $+1.1$ pp over a 10-seed same-length deep ensemble) and a clean head-to-head sweep against SWA, snapshot ensembles, and 10-seed deep ensembles across accuracy, NLL, ECE, and Brier.

The mechanism is transparent and predictive. The low-LR tail of the longer schedule drifts weights into a neighboring basin (confirmed by mode-connectivity probes and weight-space PCA), producing prediction-space diversity that weight averaging cannot reproduce. The per-sample decomposition $q \cdot g$ quantitatively predicts the ensemble gain ($0.131 \times 0.103 = 1.35$ pp, observed 1.50 pp), and the gain scales as a power law in the headroom $(1 - p_{\text{ind}})$ with $\alpha = 0.85$—so the trick pays off most on exactly the capacity-bound tasks where ensembling is most needed.

A single integer change in the training schedule, a reliable gain beyond a true 10-seed deep ensemble, and a predictive scaling law that tells you in advance when to expect the gain: TRAJLENENS is as close to a "free lunch" as deep ensembling currently offers. We expect it to compose cleanly with any diversity axis that future work identifies.

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Supplementary Material

S1 Full EMOD d6 SOTA recipe

Our FACED SOTA uses the following recipe (inherited from prior mechanism work):

- **Backbone.** EMOD axial-transformer ([13]).
- **Depth extension.** Pretrained EMOD has 3 axial layers; we increase `transformer_depth=6` and load the pretrained checkpoint with `strict=False`, keeping layers 0–2 pretrained and randomly initializing layers 3–5 ("progressive stacking" of [21]).
- **Augmentation.** CBraMod-style 4-op augmentation (jitter / noise / channel-dropout / amplitude-scale) with $p_{\text{aug}} = 0.6$.
- **Label smoothing.** $\text{LS} = 0.1$ on the cross-entropy loss.
- **Knowledge distillation.** Auxiliary KD term with $\lambda = 0.1$, $\tau = 0.1$ and a fixed random orthonormal 9×1024 prototype tensor (we showed earlier that replacing this with LLM-derived vectors does not change the result within noise).
- **Optimizer.** Adam with $\text{lr} = 10^{-3}$, weight decay 10^{-4}, backbone-LR multiplier 0.5.
- **Schedule.** Cosine annealing with $T_{\max} = 100$ (group A) or $T_{\max} = 150$ (group B), $\text{lr}_{\min} = 10^{-6}$.
- **Batch size.** 64.
- **Seeds.** $\{2025, 42, 123, 456, 789\}$ for each group.

S2 Full per-seed results

Table 10: Per-seed test BAcc for FACED d6 100ep and 150ep trainings.

Seed	100ep BAcc	150ep BAcc
2025	0.6638	0.6565
42	0.6549	0.6526
123	0.6479	0.6573
456	0.6625	0.6486
789	0.6612	0.6755
mean	0.6581	0.6581

S3 Scheduler ablation

Does the trajectory-length ensemble trick depend on cosine annealing specifically? We repeat the CIFAR-10 mixed-ensemble test (5 seeds $\times$ 50ep + 5 seeds $\times$ 75ep, same small CNN) with four different learning-rate schedules:

Table 11: Scheduler ablation on CIFAR-10. The mixed-ensemble gain over the A-only 5-seed ensemble is positive for all four schedulers, showing that the trick is not cosine-specific.

Scheduler	A mean	A ens	B ens	Mixed (+ over A ens)
Cosine	0.8878	0.9041	0.9166	0.9124 ($+0.0083$)
Step (0.1 at 50%, 75%)	0.8855	0.9004	0.9124	0.9091 ($+0.0087$)
Linear decay	0.8892	0.9042	0.9161	0.9125 ($+0.0083$)
Constant LR	0.8658	0.9043	0.9058	0.9113 ($+0.0070$)

Findings. (i) All four schedulers produce a positive mixed-ensemble gain of $+0.0070$–$+0.0087$ over the 5-seed same-length ensemble. (ii) Constant LR gives the *smallest* individual accuracy (0.866) but still benefits from the trick; the proportional single-group ensemble gain ($+0.039$) is much larger, consistent with our gain-vs-headroom scaling law. (iii) Cosine and linear decay give essentially identical results (0.9124 vs. 0.9125), so the only ingredient is "some decay from high to low LR" — the specific curve shape barely matters.

S4 Ensemble size (N) ablation

We trained 10 seeds at each of $T_A = 50$ and $T_B = 75$ epochs on CIFAR-10 (small CNN) and computed ensemble accuracy for $N \in \{1, \ldots, 10\}$ for three configurations: A-only (N seeds at 50ep), B-only (N seeds at 75ep), and mixed (N-from-A + N-from-B, $2N$ total).

Table 12: N-seed ablation on CIFAR-10 small CNN. All numbers are test accuracy on the CIFAR-10 test set.

N	A-only ($N\times$50ep)	B-only ($N\times$75ep)	Mixed ($N\times$50ep + $N\times$75ep)
1	0.8890	0.8982	0.9042
2	0.8993	0.9102	0.9094
3	0.9040	0.9122	0.9112
4	0.9040	0.9147	0.9129
5	0.9060	0.9150	0.9141
6	0.9075	0.9169	0.9153
7	0.9077	0.9187	0.9159
8	0.9085	0.9185	0.9167
9	0.9082	0.9190	0.9156
10	0.9086	0.9185	0.9166

Findings. All three configurations saturate around $N = 8$. On this particular trainer, B-only (longer schedule) plateaus slightly higher than mixed (0.919 vs. 0.917), because 75-epoch training has $+1$ pp absolute advantage over 50-epoch and the marginal gain from trajectory-length diversity is smaller than on the FACED flagship. On harder capacity-bound tasks (e.g. FACED) the mixed ensemble consistently beats same-length, consistent with our gain-vs-headroom scaling law. $N = 5$ (the paper's default) achieves 97–99% of the $N = 10$ saturation on all three settings.

S5 Alternative endpoints

We ran small ablations varying T_B/T_A. Results on CIFAR-10:

Table 13: Effect of T_B on CIFAR-10 mixed-length ensemble gain (with $T_A = 50$ fixed).

T_B	5-seed T_A ens.	Mixed ensemble	Gain over T_A ens.
50 (same, control)	0.9209	0.9209	0.000
65	TBD	TBD	TBD
75 (main)	0.9209	0.9253	+0.0044
100	TBD	TBD	TBD

S6 Training logs for 100ep vs 150ep FACED d6

Full per-epoch val BACC curves are available in the supplementary data release (`training_log_analysis.json`). The key observation is that the 100ep and 150ep trajectories reach nearly identical val BACC by their respective final epochs (0.6771 vs. 0.6776), but the 150ep trajectory is behind at epoch 99 (0.6722) because the cosine schedule still has $\sim 25\%$ of the base LR remaining.

S7 Error-correlation analysis

Table 14: Error correlation (Cohen κ on error indicators) within and between the 100ep and 150ep groups. Lower κ means more independent errors, which drives ensemble gain.

Comparison	Cohen κ
Within 100ep group (10 pairs)	0.6935
Within 150ep group (10 pairs)	0.6786
Cross-group 100ep $\times$ 150ep (25 pairs)	0.7024